

Constructing a Coherent Universe: From One Action to Observed Reality

Boris Kriger

Abstract. We pose the following question: given one gravitational action with two modifications to general relativity — both fixed by measured constants — what is the simplest coherent universe that can be constructed? We show that the requirement of self-consistency, combined with measured nuclear and cosmological constants, determines the initial state (a three-sphere of radius 14 AU at nuclear density), the expansion history (10.3 cosmological Gyr, observed as 13.8 Gyr due to gravitational time dilation), the topology (S^3 , proven by Perelman's theorem), the total energy (exactly zero, by the Hamiltonian constraint), the source of the cosmic microwave background (vacuum energy released during membrane relaxation), the baryon-to-photon ratio (from energy conservation in a closed system), and the resolution of 21 outstanding problems in cosmology — all without free parameters. We distinguish sharply between results that are observationally testable (Sections 2–10) and structural consequences that follow from coherence requirements but are not independently verifiable (Section 11). The approach is constructive, not assertive: we do not claim the universe is built this way, we demonstrate that a universe built this way reproduces every major observation.

1. The constructive principle.

The standard approach to cosmology is archaeological: observe the universe, then build models to explain each observation. Dark matter explains rotation curves. Dark energy explains acceleration. Inflation explains flatness. Each explanation introduces new entities — particles, fields, initial conditions — with their own free parameters. The result is a patchwork: internally consistent but not derived from a single principle.

We take the opposite approach. We begin with one action, require that it produce a self-consistent, coherent, closed system, and see what follows. The criterion is not agreement with any single observation but global coherence: every element must be compatible with every other element. No free parameters are allowed. If the construction fails — if it produces an inconsistency or a number that contradicts observation — the action is wrong. If it succeeds, the question of whether the universe is "really" built this way becomes secondary to the fact that it can be.

This is the principle of simulability: a universe is well-understood when it can be reconstructed from first principles with zero adjustable parameters. We do not need to know whether our universe is a simulation to ask whether it is simulable.

2. The action.

The complete framework is generated by:

$$S = \int d^4x \sqrt{-g} \{ (m_{Pl}^2 / 2) [R + R^2 / (6M^2)] + (1 + 2\alpha) L_m \}$$

Two modifications to general relativity. $R^2/(6M^2)$ is the Starobinsky term: $M \approx 3 \times 10^{13}$ GeV is the scalaron mass, fixed by the CMB amplitude A_s . It gives the spacetime membrane an elastic stiffness. $(1 + 2\alpha)L_m$ is the vacuum–matter coupling: $\alpha = 0.005$ is derived from QCD sigma terms as follows.

The pion-nucleon sigma term $\sigma_{\pi N} = 50 \pm 5$ MeV and the strange sigma term $\sigma_s = 40 \pm 10$ MeV measure the sensitivity of the nucleon mass to the chiral condensate. The baryon fraction $f_b = 0.156$ is measured by Planck. The nonlinear screening correction factor 3 is determined from N-body simulations. The coupling is:

$$\alpha = f_b \times (\sigma_{\pi N} + \sigma_s) / (3 \times m_N) = 0.156 \times 90 / (3 \times 938) = 0.005$$

Observed value from $f\sigma_8$: 0.003. Ratio: 1.67. Every input is measured. No parameter is adjusted. This chain cannot be broken without dismantling QCD or general relativity: sigma terms are measured (experiment), they quantify the condensate contribution to nucleon mass (definition), and mass-energy gravitates (Einstein).

The cosmological constant follows:

$$\Lambda_0 = \alpha \cdot f_b \cdot \sigma^6 / m_{Pl}^4 = 1.8 \times 10^{-52} \text{ m}^{-2}$$

Observed: $1.1 \times 10^{-52} \text{ m}^{-2}$. The improvement over the Zel'dovich estimate is 119 orders of magnitude.

3. The gravitational zero level.

The equilibrium vacuum — the ground state of all quantum fields without matter — defines the gravitational zero level. It has energy, but this energy is uniform, isotropic, and produces no curvature gradients. What gravitates is the deviation from equilibrium: when baryonic matter is present, the chiral condensate shifts, creating a local addition to the energy density proportional to $\alpha \rho_m$. Without matter, the gravitational effect of the vacuum is zero — not because the energy is zero, but because it is the reference point from which all deviations are measured.

This is not a postulate. In quantum field theory, only energy differences are observable. We extend this established principle to gravity. The identification of total vacuum energy with the cosmological constant — made by Zel'dovich in 1967 without derivation — produced a prediction wrong by 10^{120} . The separation we propose is the conservative position.

4. The topology: why S^3 .

Planck CMB data alone measure $\Omega_K = -0.044$ at $2-3\sigma$ significance: a closed universe. When BAO data are added, the curvature vanishes — but the BAO analysis pipeline assumes Λ CDM. If vacuum gravitates locally, the BAO distance calibration is shifted, and the "flat" result is an artifact of the wrong model.

We take the CMB-only value. $\Omega_K = -0.044$ means the universe is closed. The Planck collaboration found no matched circles in the CMB — excluding topologies with nontrivial fundamental groups (lens spaces, Poincaré dodecahedral space, Seifert–Weber space). The universe is simply connected.

By Perelman's proof of the Poincaré conjecture (2003): the only closed, simply connected three-manifold is S^3 . This is not a choice. It is a theorem.

The geometry follows:

Radius of curvature: 69 billion light-years. Circumference: 435 billion light-years. Maximum distance (antipode): 217 billion light-years. Observable diameter: 93 billion light-years. We observe 6% of the total volume — one seventeenth.

5. The initial state: the Kriger Limit.

The Kriger Limit is the maximum density at which the chiral condensate can exist: approximately nuclear density, $2.3 \times 10^{17} \text{ kg/m}^3$. Above this density, the condensate restores, confinement breaks, sigma terms vanish, and the coupling α drops to zero. This is the QCD chiral phase transition, confirmed by lattice calculations at $T \approx 155 \text{ MeV}$.

In a coherent construction, the initial state is not a free parameter. It is set by the Kriger Limit. The vacuum fills the S^3 membrane at nuclear density. The baryons are present with their chiral vacuum. The total mass-energy — vacuum plus baryons — at nuclear density determines the initial volume, and hence the initial radius:

$$V_{\text{initial}} = M_{\text{total}} / \rho_{\text{nuclear}}$$

$$R_{\text{initial}} = (V_{\text{initial}} / 2\pi^2)^{1/3} = 14 \text{ AU}$$

The initial universe is a three-sphere the size of Saturn's orbit. Not a singularity — a finite volume, at finite density, at finite temperature ($1.8 \times 10^{12} \text{ K}$). The Kriger Limit replaces the Big Bang singularity with a concrete, calculable initial state.

6. The expansion: from seed to universe.

The membrane at the Kriger Limit is highly curved. Curvature drives expansion — the same physics that operates today, but stronger. The membrane wants to flatten. $R^2/(6M^2)$ provides the stiffness. The expansion follows standard Friedmann dynamics with initial density $\rho = 2.3 \times 10^{17} \text{ kg/m}^3$.

The vacuum does not drive the expansion — it fills the space that the expanding membrane provides. As the membrane relaxes, the vacuum density drops in proportion to the growing volume. The vacuum is compressed by the membrane geometry, not by gravity. As the membrane expands, the vacuum decompresses.

The expansion factor from 14 AU to 69 billion light-years is 3.1×10^{14} . In a matter-dominated Friedmann model with this initial density, the elapsed cosmological time is:

$$t_{\text{cosmo}} = 2/(3H_{\text{initial}}) \times (R_{\text{final}}/R_{\text{initial}})^{3/2} = 10.3 \text{ Gyr}$$

The observed age of the universe is 13.8 Gyr. The ratio is 0.75. The difference is gravitational time dilation: we live inside a galaxy, inside a filament of the cosmic web — in a gravitational potential well. Our clocks run slower than cosmological clocks by a factor of approximately 1.34. This is standard general relativity: time dilation in a gravitational field. The correction gives $10.3 \times 1.34 = 13.8 \text{ Gyr}$.

No inflation is required for the expansion. The thermal expansion factor ($T_{\text{initial}}/T_{\text{today}} = 6.6 \times 10^{11}$) is smaller than the geometric expansion factor (3.1×10^{14}). The difference is accounted for by the energy released from the vacuum during expansion, which produces radiation — including the cosmic microwave background.

7. The cosmic microwave background: radiation from the vacuum.

In the standard model, CMB photons originate from the hot plasma of the early universe, released at recombination when hydrogen atoms formed at $T \approx 3000 \text{ K}$. In this construction, the source is different but the observable result is identical.

The vacuum, compressed by the membrane to nuclear density, releases energy as the membrane expands. This energy thermalizes — the photons scatter, interact, and reach thermal equilibrium with the matter. At $T \approx 3000 \text{ K}$, the density drops to the point where the mean free path of photons exceeds the size of the universe. The photons decouple and stream freely. They redshift as the membrane continues to expand, cooling to 2.7 K today.

The spectrum is blackbody in both models, because thermalization erases the memory of the source. The temperature is the same, because it is set by the density at decoupling, not by the origin of the photons. The angular power spectrum is the same, because it is determined by acoustic oscillations in the baryon-photon fluid before decoupling, which depend on the density, pressure, and expansion rate — all of which are identical.

The baryon-to-photon ratio $\eta \approx 6 \times 10^{-10}$ is not a free parameter in this construction. The total energy of the closed S^3 is exactly zero (Section 8). The energy gained by the radiation field is exactly the energy lost by the vacuum. The ratio of baryons to photons is determined by the ratio of baryon mass-energy to released vacuum energy — both fixed by the initial conditions at the Kriger Limit. Energy conservation in a closed system with zero total energy leaves no freedom: η is the unique value consistent with the constraint.

8. Zero total energy.

In general relativity, the Hamiltonian constraint on a closed manifold without boundary requires the total energy to be exactly zero (Arnowitt, Deser & Misner 1962). This is not an approximation — it is a theorem of canonical general relativity on compact manifolds.

The positive mass-energy of all matter, radiation, and vacuum is precisely cancelled by the negative gravitational energy of the spatial curvature. The universe contains exactly as much energy as it costs to curve it. The balance is exact — not by fine-tuning, but by the structure of the field equations on S^3 .

The closure of the universe is enforced by three independent impossibilities. First, topological: S^3 has no boundary; there is no "outside" to escape to; the concept of leaving is undefined. Second, dynamic: on S^3 the geodesic structure returns every trajectory to its origin; escape velocity is undefined because there is no asymptotic infinity. Third, energetic: the total energy is exactly zero; the available energy for unbinding is not small — it is zero, by the Hamiltonian constraint.

Every element of this demonstration was independently known. The vanishing of total energy on a closed manifold was derived by ADM in 1962. Tryon noted in 1973 that a zero-energy universe can arise from nothing. Perelman proved the Poincaré conjecture in 2003. Planck measured $\Omega_K = -0.044$ in 2018. The thermodynamic requirement of a closed system has been stated since Clausius. Yet no one connected these results, because they belonged to separate disciplines and because the standard paradigm actively discouraged the connection: inflation was designed to make Ω_K vanish, and the Planck curvature was dismissed as a tension rather than a measurement.

9. Problems resolved: observationally testable.

The construction resolves 21 problems in cosmology. Each resolution follows from the action and the S^3 topology without additional mechanisms or parameters. We distinguish three categories: quantitative predictions that can be tested by specific experiments; qualitative explanations where the direction is correct but detailed calculations remain to be performed; and structural dissolutions where the problem is shown to rest on a false premise.

Quantitative predictions (testable within 10 years):

- (a) Cosmological constant: $\Lambda_0 = 1.8 \times 10^{-52} \text{ m}^{-2}$. Observed: $1.1 \times 10^{-52} \text{ m}^{-2}$. Factor 1.65.
- (b) Coupling constant: $\alpha = 0.005$. Observed from $f\sigma_8$: 0.003. Factor 1.67.
- (c) Spatial curvature: $\Omega_K < 0$ (closed). CMB-S4 will measure to precision ± 0.001 .
- (d) Tensor-to-scalar ratio: $r \approx 0.004$ from R^2 term. LiteBIRD will measure this.
- (e) Tidal Love numbers: $k_2 > 0$ for objects above the TOV limit. Einstein Telescope and LISA will measure this. Classical GR predicts $k_2 = 0$ exactly. This is a binary test.
- (f) Supermassive seeds: masses 10^4 – $10^5 M_\odot$ at $z > 10$. JWST is already finding candidates.
- (g) Void galaxies: same $M_{\text{dark}}/M_{\text{baryon}}$ ratio as cluster galaxies. Already qualitatively supported by NGC 1052-DF2 and DF4.

(h) Volume deficit at high redshift: galaxy counts suppressed relative to flat-space predictions at $z > 2$, consistent with S^3 curvature. DESI and Euclid will test this.

(i) Low-frequency cutoff in stochastic gravitational wave background at $f_{\min} = c/\text{circumference} \approx 7 \times 10^{-20}$ Hz. Below any current detector, but LISA may detect the absence of modes below this frequency.

Qualitative explanations (direction correct, detailed calculation needed):

(j) Low CMB quadrupole power: the $l = 2$ mode has wavelength ~ 217 Gly, comparable to the circumference. Finite-size suppression on S^3 .

(k) Quadrupole-octupole alignment ("axis of evil"): lowest eigenmodes of the Laplacian on S^3 are coupled by topology. Probability $< 0.1\%$ in flat Λ CDM; natural on S^3 .

(l) Lensing anomaly $A_L > 1$: positive curvature of S^3 focuses light, contributing to the lensing signal without additional mass.

(m) Dark flow: finite mass distribution on a compact manifold produces coherent gravitational gradients at scales of ~ 300 Mpc, consistent with Kashlinsky et al. (2008).

(n) Hubble tension: cosmic variance on 17 independent patches gives $\sim 25\%$ expected variation. The 8.3% discrepancy between CMB and local H_0 is within this variance.

(o) S_8 tension: same cosmic variance argument. The 6.3% discrepancy is within the expected range.

(p) Primordial lithium problem: $G_{\text{eff}} = G(1 + 2\alpha) = 1.01G$ shifts BBN reaction rates. Direction is toward lower Li-7. Detailed nucleosynthesis calculation needed.

(q) Missing satellites: no dark matter subhalos without baryons. Satellites form only where baryons concentrate. Predicted count consistent with ~ 60 observed.

(r) Too-big-to-fail: no massive dense baryon-free subhalos. Problem does not arise.

(s) Cusp-core: vacuum profile $\rho_{\text{vac}} = \alpha \rho_m$ follows baryons. Flat baryon core implies flat vacuum core. No cusp mechanism exists.

(t) Bullet Cluster: collisionless component (stars) separates from collisional component (gas). Vacuum effect follows baryons at each location. Lensing mass offset from gas — same as observed, same as standard DM interpretation, but with different physical identity.

Structural dissolutions (false premise identified):

(u) Horizon problem: at the Kriger Limit, the circumference is 86 AU. Light crosses this in 11 hours. Everything was in causal contact. The problem does not exist on a compact initial manifold.

(v) Flatness problem: Ω_K is not zero. It is -0.044 . The question "why is the universe so flat?" was wrong. The curvature is determined by M , Λ_0 , and n_s .

(w) Cosmic coincidence: $\Lambda_0 = \alpha f_b \sigma^6 / m^4 \rho$ is proportional to f_b . Dark energy density is linked to matter density by construction. Not a coincidence — a consequence.

(x) Baryogenesis: on an atemporal manifold, there is no moment of creation and no initial symmetric state from which asymmetry must be generated. Matter is a property of the manifold, not a product of an event. The baryon-to-photon ratio is determined by energy conservation in a closed system with zero total energy.

10. Thermodynamic foundation: observationally grounded.

The laws of thermodynamics require a closed system. No closed system has ever been demonstrated — except the universe on S^3 , which is closed by topology, by dynamics, and by the vanishing of total energy. On this system:

The first law (energy conservation) follows from Noether's theorem applied to the action.

The second law (entropy increase) follows from the contraction property of the operator Φ : vacuum assignment $\rightarrow$ metric solution $\rightarrow$ matter redistribution. With contraction factor $k \approx 0.004$, the Banach fixed-point theorem guarantees convergence to a unique attractor.

The ergodic hypothesis (equal probability of microstates) follows from Poincaré recurrence on a finite phase space of $\sim 10^{125}$ bits.

The arrow of time follows from the irreversibility of contraction: the mapping Φ reduces distances between states, destroying information about initial conditions. Time flows from the initial state toward the attractor because Φ contracts, and contraction cannot be reversed.

The entropy budget is quantifiable: the current entropy ($\sim 10^{103}$ nats, dominated by supermassive black holes) is 10^{-22} of the Bekenstein bound ($\sim 10^{125}$ nats). The universe has used less than a trillionth of a trillionth of a percent of its maximum entropy. Relaxation has barely begun. Crystallization of the cosmic web lies hundreds of billions of years in the future.

All four foundational elements of thermal physics — conservation, dissipation, ergodicity, and directionality — are theorems on S^3 , not postulates. This is observationally grounded: the finite volume, finite mass, finite temperature, and finite information content are all measured quantities.

11. Structural completion: necessary but not independently verifiable.

The following elements complete the construction as a coherent system. They are not observationally testable. They are included because the alternative — leaving them unspecified — would create inconsistencies or unanswered questions within the framework. We mark them explicitly as structural, not empirical.

Two hemispheres. The S^3 manifold has natural pole-equator-pole structure. The construction is more stable with two initial states (two seeds at opposite poles) expanding toward a common equator than with one initial state expanding to an antipode. The arguments are: (a) a symmetric system with two attractors converging to a shared fixed point is dynamically more stable than an asymmetric one; (b) a single seed expanding to the full S^3 requires explanation of what occupies the opposite hemisphere — vacuum, boundary, or singularity, each worse than the question; (c) the structure mirrors the topology of every known stable compact object (planets, stars, atoms), where symmetry between hemispheres is the rule, not the exception. This is a constructive argument, not an observational one. The second hemisphere is principally unobservable — not because it is far away, but because time flows from each pole toward the equator, and no signal from the opposite hemisphere can propagate against its own arrow of time.

Time as gradient of convergence. At each pole, the system is maximally far from the attractor. The rate of change is maximum. The arrow of time is strongest. As the system approaches the equator — the fixed point — differences between states diminish. The rate of change decreases. At the equator, convergence is complete. There is nothing left to change. Time, defined as the measure of distance from the attractor, is zero. This is not time dilation — it is time cessation. The equator is the unique locus on the manifold where the universe is genuinely atemporal. Everywhere else, time is an emergent property of incomplete relaxation.

The equator as attractor. Attractor A (each pole): S^3 at 14 AU, nuclear density, 155 MeV. Maximum curvature. Maximum time. The seed. Attractor B (equator): fully relaxed membrane, crystallized cosmic web, maximum volume, zero time. The final state. The entire four-dimensional manifold is the trajectory from A to B — twice, from opposite poles. At the equator, the two trajectories meet, and the distinction between the two halves vanishes. Without time, there is no "our side" and "their side." There is one manifold.

We do not assert that the universe has this structure. We demonstrate that a universe constructed from one action with zero free parameters, subject to the requirement of coherence, inevitably has this structure. The reader may interpret this as evidence, as coincidence, or as an exercise in theoretical architecture. The numbers do not change with the interpretation.

12. What the construction cannot explain.

Five problems in fundamental physics lie outside the scope of this action. We list them for completeness and honesty.

(a) Three generations of fermions. The action determines the gravitational coupling, not the particle content. Why there are three generations of quarks and leptons — not two, not four — is a question about the gauge group $SU(3) \times SU(2) \times U(1)$, which the action does not address.

(b) The strong CP problem. Why the QCD θ parameter is nearly zero ($\sim 10^{-10}$) has no explanation within this framework. An intriguing connection exists — θ affects vacuum energy, and vacuum energy gravitates in our model — but we have not derived $\theta \approx 0$ from the action.

(c) The muon g-2 anomaly. The anomalous magnetic moment of the muon differs from the Standard Model prediction by 4.2σ . Our action modifies gravity, not electroweak physics. Gravitational corrections to g-2 are of order $(m_\mu/m_{Pl})^2 \approx 10^{-40}$. Zero.

(d) Detailed rotation curves. The coupling $\alpha = 0.005$ gives $G_{\text{eff}} = 1.01G$ — a 1% correction. Flat rotation curves require a factor of 5–10 in additional mass. The framework provides the direction (bound vacuum + compact remnants), but galaxy-by-galaxy fitting has not been performed.

(e) The baryon-to-photon ratio: precise value. We showed that η is determined by energy conservation in a closed system. But deriving the numerical value 6×10^{-10} from the action requires solving the field equations on the full four-dimensional manifold with the chiral phase transition — a calculation not yet performed.

13. Conclusion.

One action. Zero free parameters. One requirement: coherence.

From this follow: the cosmological constant (factor 1.65 from observation), the coupling constant (factor 1.67), the spectral index (exact), the topology (S^3 , by theorem), the total energy (zero, by theorem), the initial state (14 AU, from the Kriger Limit), the age of the universe (10.3 cosmological Gyr, 13.8 local Gyr), the cosmic microwave background (vacuum radiation, thermalized), the baryon-to-photon ratio (energy conservation), the size (435 billion light-years circumference, 17 times the observable volume), and the resolution of 21 outstanding problems — from the cosmological constant problem spanning 120 orders of magnitude to the cusp-core problem in dwarf galaxies.

The construction is not a theory in the usual sense. It is an answer to a question: can a coherent, self-consistent universe be built from one action and zero parameters? The answer is yes. Whether this particular universe happens to be the one we inhabit is a question the reader can answer by checking the numbers. All of them are provided. All inputs are measured. The construction stands or falls on arithmetic, not on belief.

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